

Assignment - 01

Analogy & Mathematical Modeling of Boost and Buck-Boost Converters

Introduction

Both Boost and Buck-Boost converters are fundamental DC-DC power converter topologies. They utilize the exact same four core components: an inductor (L), a controlled semiconductor switch (S), a diode (D), and an output capacitor (C). The primary distinction between the two lies purely in their circuit topology—specifically, how these elements are arranged relative to the source and the load.

1. Basic Circuit Configuration & Simulation

Boost Converter

The Boost converter acts as a step-up converter. The inductor is located at the input side, leading to continuous input current.

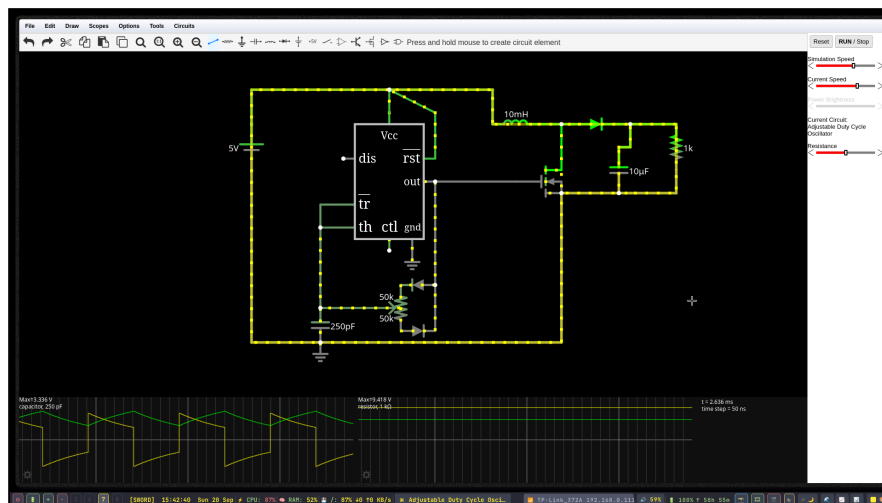
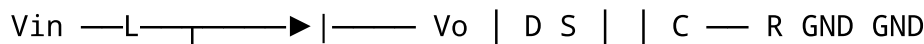


Figure 1: Boost Converter Simulation (using 555 Timer PWM configuration)

As observed in the simulation (Figure 1), the switching action of the MOSFET causes the inductor to charge during the ON-state and discharge into the output capacitor during the OFF-state, producing a voltage greater than the input.

Buck-Boost Converter

The Buck-Boost converter can both step up and step down the input voltage. Its topology features the switch in series with the source, and the inductor placed in parallel with the load. This arrangement results in an inverted output voltage polarity.

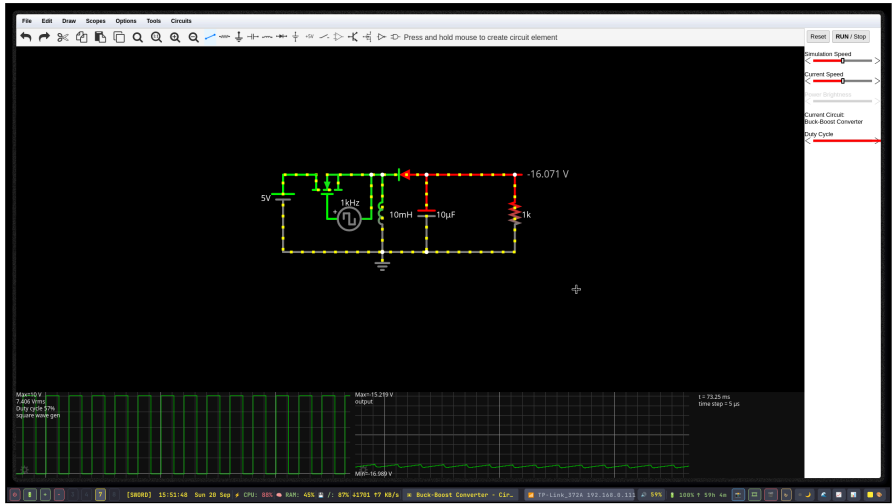
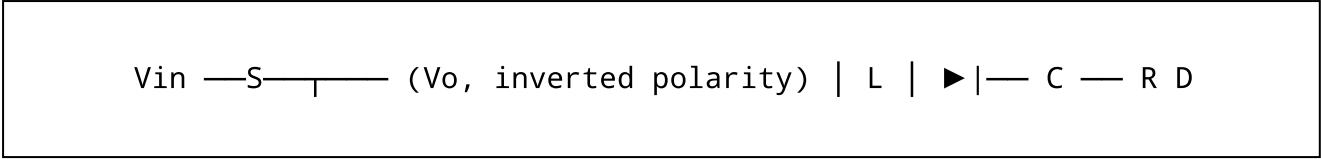


Figure 2: Buck-Boost Converter Simulation (Demonstrating inverted negative output voltage)

Figure 2 verifies the characteristic inverting nature of the Buck-Boost topology, where a positive input generates a regulated negative output voltage relative to a common ground.

2. Analogy Between Boost and Buck-Boost

Feature	Boost Converter	Buck-Boost Converter
Output Polarity	Same as input (Positive)	Inverted (Negative)
Output Magnitude	Always $\geq V_{in}$ (Step-up only)	Can be $<$ or $> V_{in}$ (Step up/down)
Inductor during OFF-state	Connected directly to load (direct energy transfer)	Isolated from source, dumps into load (indirect/energy-transfer type)
Voltage Gain Formula	$V_o / V_{in} = 1 / (1 - D)$	$V_o / V_{in} = D / (1 - D)$

Key Mathematical Analogy: The Buck-Boost transfer function is conceptually the mathematical product of the Buck converter gain and the Boost converter gain:

3. Mathematical Model — Boost Converter

The system is modeled by analyzing the two distinct topological states during one switching period T , governed by the duty cycle D .

Switch ON ($0 \rightarrow DT$): The inductor charges from the source; the diode is reverse biased; the capacitor alone supplies the load.

$$L \frac{di_L}{dt} = V_{in} \quad C \frac{dv_C}{dt} = -\frac{v_C}{R}$$

Switch OFF ($DT \rightarrow T$): The inductor discharges through the diode into the load.

$$L \frac{di_L}{dt} = V_{in} - v_C \quad C \frac{dv_C}{dt} = i_L - \frac{v_C}{R}$$

State-Space Averaged Model (weighted by d and $1-d$):

$$L \frac{di_L}{dt} = V_{in} - (1-d)v_C$$

$$C \frac{dv_C}{dt} = (1-d)i_L - \frac{v_C}{R}$$

Steady-State Solution (Applying volt-second and charge balance, derivatives = 0):

$$\boxed{\frac{V_o}{V_{in}} = \frac{1}{1-D}} \quad I_L = \frac{V_o}{R(1-D)}$$

4. Mathematical Model — Buck-Boost Converter

Switch ON ($0 \rightarrow DT$): Inductor charges directly from V_{in} ; diode blocks; capacitor supplies load.

$$L \frac{di_L}{dt} = V_{in} \quad C \frac{dv_C}{dt} = -\frac{v_C}{R}$$

Switch OFF ($DT \rightarrow T$): Inductor discharges through diode into the (inverted-polarity) output.

$$L \frac{di_L}{dt} = -v_C \quad C \frac{dv_C}{dt} = i_L - \frac{v_C}{R}$$

State-Space Averaged Model:

5. Solar (PV) Plant Application — Why Buck-Boost is Preferred

The Problem: A Photovoltaic (PV) panel's terminal voltage (V_{pv}) swings significantly with changes in solar irradiance and temperature. Depending on environmental conditions, V_{pv} may rise above or drop below the required battery or DC-bus voltage. A simple Buck or simple Boost converter cannot handle bidirectional voltage regulation. The Buck-Boost converter naturally accommodates both scenarios.

PV Source Model (Single-Diode Equivalent):

$$I_{pv} = I_{ph} - I_0 \left[\exp \left(\frac{q(V_{pv} + I_{pv}R_s)}{nkT} \right) - 1 \right] - \frac{V_{pv} + I_{pv}R_s}{R_{sh}}$$

Coupling the PV source to the Buck-Boost converter (MPPT interface):

Assuming lossless power conversion ($P_{in} = P_{out}$), we know:

$$V_{pv} \cdot I_{pv} = \frac{V_o^2}{R_{load}}$$

Since $V_o = [D / (1 - D)] \cdot V_{pv}$, the effective input resistance (R_{in}) seen by the PV panel becomes:

$$R_{in} = \frac{V_{pv}}{I_{pv}} = R_{load} \cdot \left(\frac{1 - D}{D} \right)^2$$

This equation forms the core of the solar-plant MPPT (Maximum Power Point Tracking) model. By dynamically adjusting the duty cycle D (using algorithms such as Perturb & Observe or Incremental Conductance), the controller continuously reshapes the apparent input resistance R_{in} . It forces it to precisely match the PV panel's optimum resistance $R_{mpp} = V_{mpp} / I_{mpp}$, ensuring the panel operates at its peak efficiency irrespective of load variations or changing weather conditions.

PV Panel(V_{pv} , I_{pv}) → Buck-Boost Converter → Battery/DC Bus (V_o)
↑ MPPT Controller (Adjusts Duty Cycle D)

6. Conclusion

While Boost and Buck-Boost converters share identical fundamental building blocks, the strategic